

MIPT & YSDA · AUTUMN 2026

# Deep Generative Models

## Lecture 6

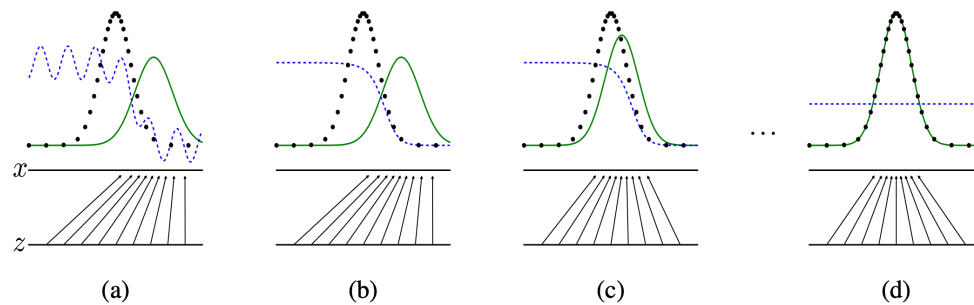
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# Recap of Previous Lecture

## Objective

$$\min_{\theta} \max_{\phi} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z}))) \right]$$



Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.

# Recap of Previous Lecture

## Standard GAN Objective

$$\min_{\theta} \max_{\phi} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z}))) \right]$$

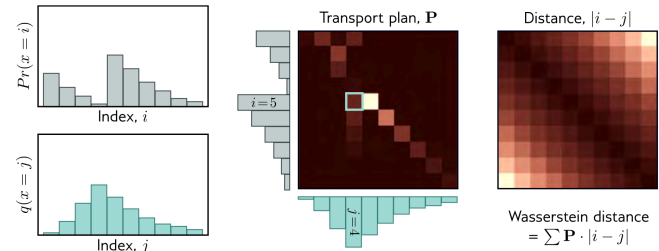
## Theoretical Results

- Both the data distribution  $p_{\text{data}}(\mathbf{x})$  and the generative distribution  $p_{\theta}(\mathbf{x})$  are supported on low-dimensional manifolds.
- If  $p_{\text{data}}(\mathbf{x})$  and  $p_{\theta}(\mathbf{x})$  are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) = D_{\text{KL}}(p_{\theta} \parallel p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \parallel p_{\theta}) = \log 2$$

# Recap of Previous Lecture



## Wasserstein Distance

$$\begin{aligned}
 W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\
 &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2
 \end{aligned}$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$  is the transport plan: the amount of “dirt” assigned from  $\mathbf{x}_1$  to  $\mathbf{x}_2$ .
- $\Gamma(\pi, p)$  denotes the set of all joint distributions  $\gamma(\mathbf{x}_1, \mathbf{x}_2)$  with marginals  $\pi$  and  $p$  ( $\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2)$ ,  $\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1)$ ).
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$  is the mass,  $\|\mathbf{x}_1 - \mathbf{x}_2\|$  is the distance.

# Recap of Previous Lecture

On a compact data space  $\mathcal{X}$ :

## Dependence on Generator Parameters

For a feedforward generator  $G_\theta$  with Lipschitz activations and  $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$ ,  $W(p_{\text{data}} \| p_\theta)$  is continuous in  $\theta$  and differentiable almost everywhere.

## Convergence of Distributions

For a fixed distribution  $\pi$  and a sequence  $p_t$  on  $\mathcal{X}$ , as  $t \rightarrow \infty$ :

$$D_{\text{KL}}(\pi \| p_t) \rightarrow 0 \implies D_{\text{JS}}(\pi \| p_t) \rightarrow 0 \implies W(\pi \| p_t) \rightarrow 0$$

The first implication also holds for  $D_{\text{KL}}(p_t \| \pi) \rightarrow 0$ ; the converses need not hold.

**Wasserstein can capture distributions getting closer even when their supports do not overlap.**

# Recap of Previous Lecture

## Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\theta}) = \frac{1}{K} \max_{\|f\|_L \leq K} [\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x})],$$

where  $f : \mathbb{R}^m \rightarrow \mathbb{R}$  is  $K$ -Lipschitz continuous ( $\|f\|_L \leq K$ ).

## WGAN Objective

$$\min_{\theta} W(p_{\text{data}} \| p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} [\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z}))].$$

- The discriminator  $D$  is replaced by function  $f$ : in WGAN, it is known as the **critic**, which is *not* a classifier.
- If the weights  $\phi$  are restricted to a compact set  $\Phi = [-c, c]^d$ , then  $f_{\phi}(\mathbf{x})$  is  $K$ -Lipschitz continuous.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\theta}) &= \max_{\|f\|_L \leq K} [\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x})] \\ &\geq \max_{\phi \in \Phi} [\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f_{\phi}(\mathbf{x})] \end{aligned}$$

# Recap of Previous Lecture

## Frechet Inception Distance (FID)

Use Inception image embeddings  $\mathbf{z} = \mathbf{f}(\mathbf{x})$ ; compare Gaussian approximations of their distributions:

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\theta}\|^2 + \text{tr} \left[ \boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\theta} - 2 \left( \boldsymbol{\Sigma}_{\text{data}}^{1/2} \boldsymbol{\Sigma}_{\theta} \boldsymbol{\Sigma}_{\text{data}}^{1/2} \right)^{1/2} \right].$$

FID depends on the feature network, Gaussian approximation and sample size; it may not correlate with human evaluation.

## CMMD: Comparing CLIP Image Embeddings

For CLIP image-embedding distributions  $P, Q$ , use a Gaussian kernel  $k$ :

$$\text{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

Here  $\mathbf{z}, \mathbf{z}' \sim P$  and  $\mathbf{w}, \mathbf{w}' \sim Q$  are mutually independent. This compares more than means and covariances.

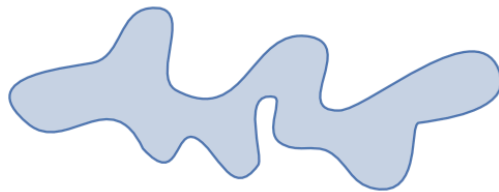
# Recap of Previous Lecture

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$  - real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$  - generated samples.

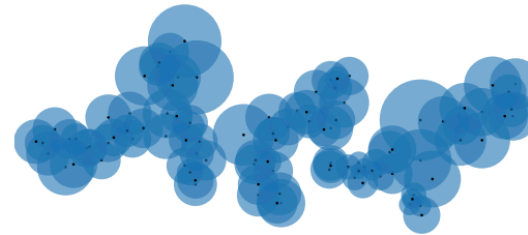
Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\theta}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\theta}).$$



(a) True manifold



(b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

# Recap of Previous Lecture

Score an image  $\mathbf{x}$  for a text prompt  $t$ .

## CLIP: Learn from Image–Caption Pairs

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)) .$$

## PickScore: Learn from Human Choices

For the same prompt, a human prefers  $\mathbf{x}^+$  to  $\mathbf{x}^-$ :

$$\mathbb{P}_{\theta}(\mathbf{x}^+ \succ \mathbf{x}^- \mid t) = \frac{\exp r_{\theta}(\mathbf{x}^+, t)}{\exp r_{\theta}(\mathbf{x}^+, t) + \exp r_{\theta}(\mathbf{x}^-, t)} .$$

Train the score  $r_{\theta}$  to assign higher probability to the human choice.

**VQAScore / GenEval / DPG-Bench:** evaluate compositional correctness; CLIP similarity can miss counts, attributes and relations.

A learned preference score can guide sample selection and model training — but optimizing the score can exploit its flaws.

# Recap of Previous Lecture

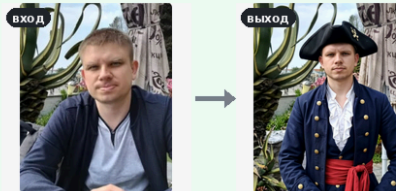
- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It's important to assess various properties.

## РЕЛЕВАНТНОСТЬ

«Выполнена ли инструкция?»

Сделал ли редактор именно то, что просили

✓ ХОРОШО



Запрос: «сделать пиратом». Образ пирата создан — то, о чём просили, выполнено.

✗ ПЛОХО



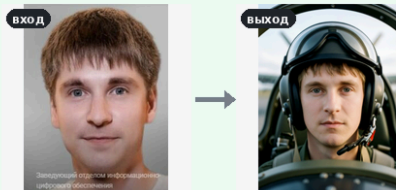
Запрос: эффект вращения на карусели. Картинка почти не изменилась — инструкция не выполнена.

## СОХРАНЕНИЕ

«Сохранилось ли то, что менять не просили?»

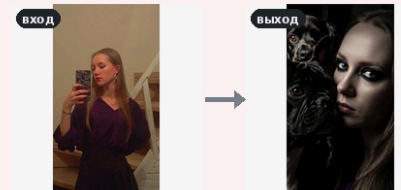
Фон, лицо и композиция остаются на месте

✓ ХОРОШО



Запрос: образ лётчика, «не меняя черты лица». Под шлемом — то же самое лицо.

✗ ПЛОХО



Запрос: «не меняй внешность, один в один», лишь приблизить лицо к собаке. Но на выходе — совершенно другой человек.

## ДЕФЕКТЫ

«Не появилось ли артефактов?»

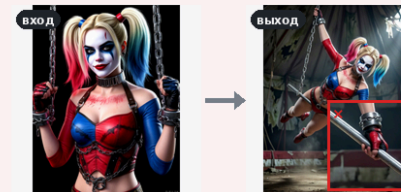
Лишние пальцы, сломанная геометрия и т. п.

✓ ХОРОШО



Кисти рук и предметы без искажений — артефактов нет.

✗ ПЛОХО



Кисть руки слилась в бесформенный ком (увеличено) — явный артефакт.

# Recap of Previous Lecture

## Unnormalized Density

$$p_{\theta}(\mathbf{x}) = \frac{\hat{p}_{\theta}(\mathbf{x})}{Z_{\theta}}, \quad \text{where } Z_{\theta} = \int \hat{p}_{\theta}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\theta}(\mathbf{x})$  can be any non-negative function.
- If we reparameterize as  $\hat{p}_{\theta}(\mathbf{x}) = \exp(-f_{\theta}(\mathbf{x}))$ , we eliminate the non-negativity constraint.

## Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\theta} = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x})$$

- Suppose we already have this density (normalized or not)  $p_{\theta}(\mathbf{x})$ .
- How can we sample from the model?

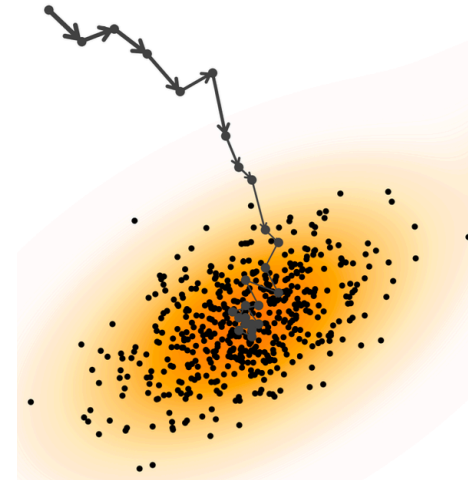
# Recap of Previous Lecture

## Theorem (Informal)

Let  $\mathbf{x}_0$  be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow  $p_\theta(\mathbf{x})$  (for sufficiently small  $\eta$  and large  $l$ ):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_\theta(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

- What if  $\boldsymbol{\epsilon}_l = \mathbf{0}$ ?
- The density  $p_\theta(\mathbf{x})$  is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to  $\mathbf{x}$ , not  $\boldsymbol{\theta}$ .
- $\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x})$  defines a vector field.



# Outline

## 01 Score Matching

Denoising Score Matching

Noise-Conditioned Score Network (NCSN)

## 02 Forward Gaussian Diffusion Process

# Outline

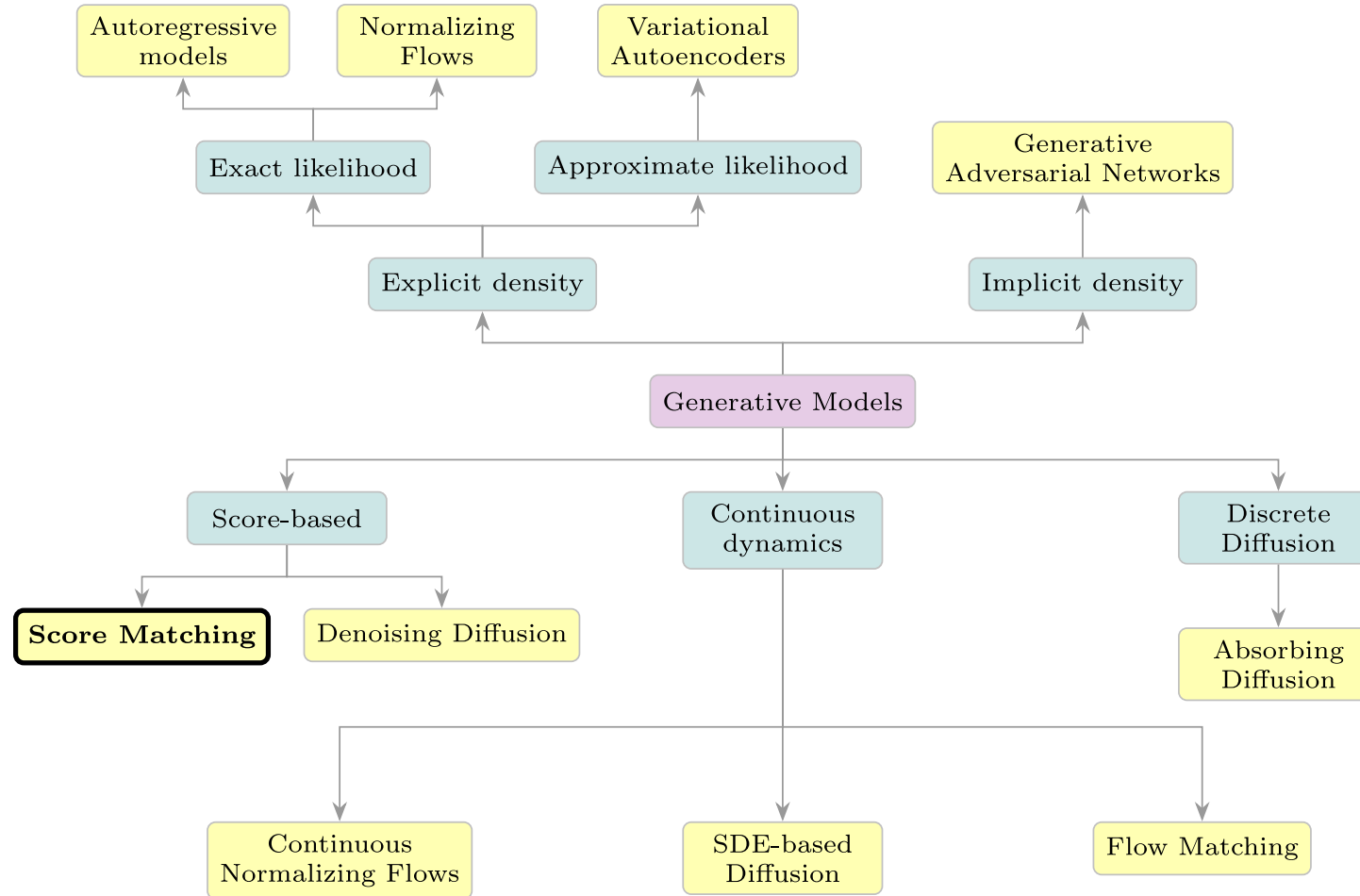
## 01 Score Matching

Denoising Score Matching

Noise-Conditioned Score Network (NCSN)

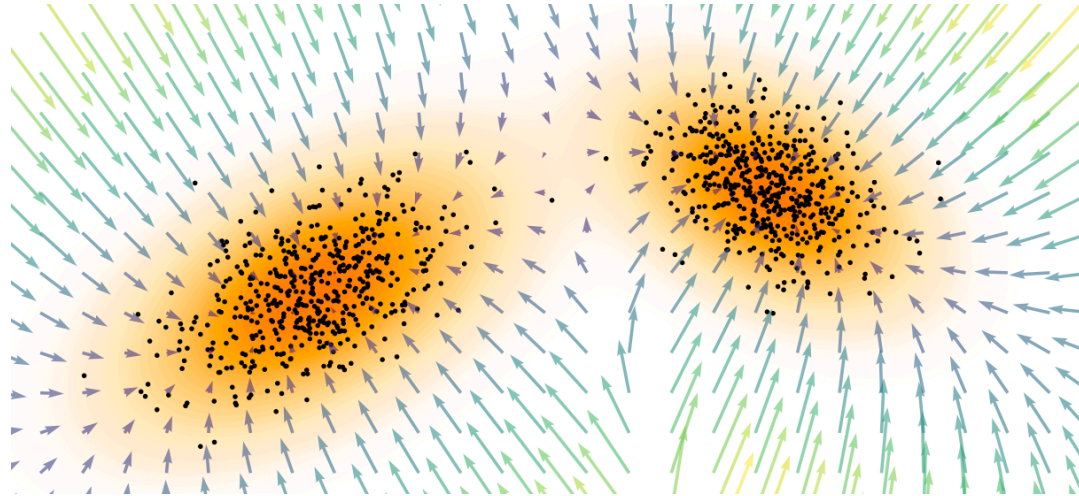
## 02 Forward Gaussian Diffusion Process

# Generative Models Taxonomy



# Score Function

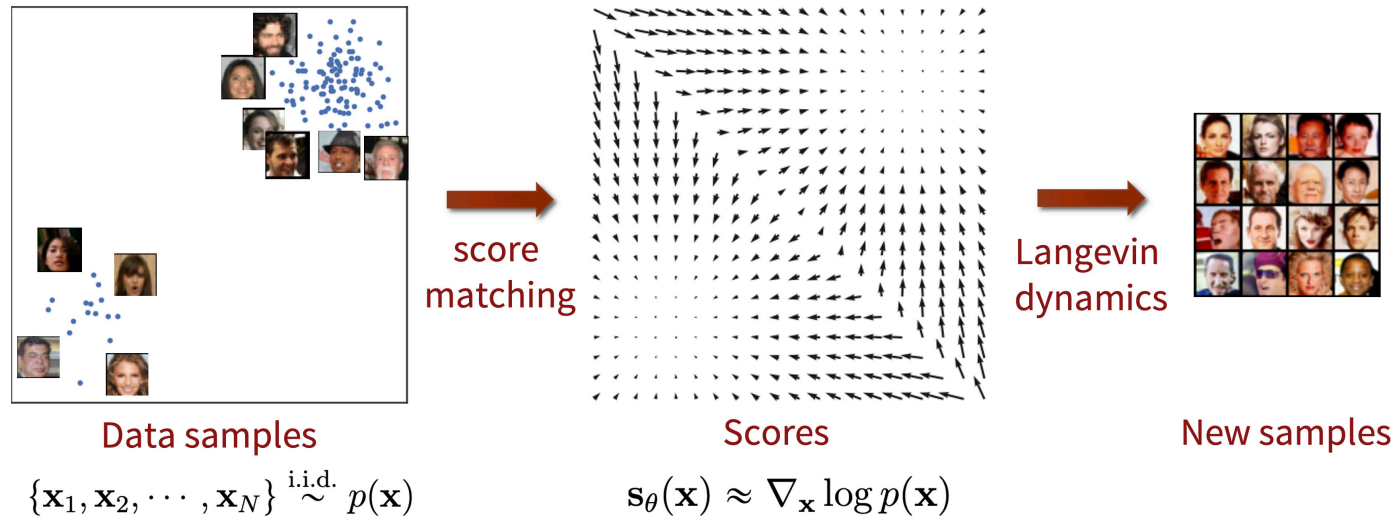
$$\mathbf{s}_\theta(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x})$$



## Langevin Dynamics

$$\begin{aligned} \mathbf{x}_{l+1} &= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\text{data}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\ &= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_\theta(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\ &= \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_\theta(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l. \end{aligned}$$

# Score Matching



## Fisher Divergence

$$\begin{aligned}
 D_F(p_{\text{data}}, p_\theta) &= \frac{1}{2} \mathbb{E}_{p_{\text{data}}} \left\| \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \\
 &= \frac{1}{2} \mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_\theta(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \rightarrow \min_{\theta}
 \end{aligned}$$

**Problem:** We don't know  $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ .

# Why Is the Score Function Better than the Density?

## Freedom from Normalization Constants

Many distributions are defined only up to an unnormalized density  $\hat{p}(\mathbf{x})$ :

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z}_{=0} = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x})$$

The score function bypasses the intractable constant  $Z$  entirely!

## A Complete Representation

The score function fully characterizes the underlying distribution (up to a constant):

$$\log p(\mathbf{x}) = \log p(\mathbf{x}_0) + \int_0^1 \mathbf{s}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))^{\top} (\mathbf{x} - \mathbf{x}_0) dt$$

Modeling the score is as expressive as modeling  $p(\mathbf{x})$  itself, while often more tractable for generative modeling.

# Outline

## 01 Score Matching

Denoising Score Matching

Noise-Conditioned Score Network (NCSN)

## 02 Forward Gaussian Diffusion Process

# Denoising Score Matching

Let us perturb the original data  $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$  with Gaussian noise:

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad q(\mathbf{x}_\sigma | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma | \mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

## Assumption

The solution to

$$\frac{1}{2} \mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \rightarrow \min_{\boldsymbol{\theta}}$$

satisfies  $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta}, 0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$  if  $\sigma$  is sufficiently small.

- The score function of the noised data nearly matches the score function of the original data.
- The score function  $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma)$  is parameterized by  $\sigma$ .
- **Note:** We don't know  $q(\mathbf{x}_\sigma)$ , just as we don't know  $p_{\text{data}}(\mathbf{x})$ .

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the following holds:

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\theta, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \\ &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma | \mathbf{x})} \left\| \mathbf{s}_{\theta, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma | \mathbf{x}) \right\|_2^2 + \text{const}(\boldsymbol{\theta}) \end{aligned}$$

## Gradient of the Noise Kernel

$$\begin{aligned} \mathbf{x}_\sigma &= \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, & q(\mathbf{x}_\sigma | \mathbf{x}) &= \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I}) \\ \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma | \mathbf{x}) &= -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma} \end{aligned}$$

- The right-hand side doesn't require computing  $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$  or even  $\nabla_{\mathbf{x}_\sigma} \log p_{\text{data}}(\mathbf{x}_\sigma)$ .
- $\mathbf{s}_{\theta, \sigma}(\mathbf{x}_\sigma)$  is trained to **denoise** the noised samples  $\mathbf{x}_\sigma$ .

# Denoising Score Matching

Initial objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \rightarrow \min_{\theta}$$

Noised objective:

$$\mathbb{E}_{q(\mathbf{x}_{\sigma})} \|\mathbf{s}_{\theta, \sigma}(\mathbf{x}_{\sigma}) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_{\sigma})\|_2^2 \rightarrow \min_{\theta}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_{\sigma}|\mathbf{x})} \|\mathbf{s}_{\theta, \sigma}(\mathbf{x}_{\sigma}) - \nabla_{\mathbf{x}_{\sigma}} \log q(\mathbf{x}_{\sigma}|\mathbf{x})\|_2^2 \rightarrow \min_{\theta}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0, \mathbf{I})} \left\| \mathbf{s}_{\theta, \sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \rightarrow \min_{\theta}$$

## Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\theta, \sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the last term is independent of  $\theta$ :

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\theta).$$

## Proof

For any integrable  $h$ , marginalize the joint distribution:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int \left( \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma).$$

Expand the squared error:

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 \\ &= \mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma)\|_2^2 + \underbrace{\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{C_1} - 2\mathbb{E}_{q(\mathbf{x}_\sigma)} [\mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)]. \end{aligned}$$

The score-norm term  $C_1$  is independent of  $\theta$ .

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the last term is independent of  $\theta$ :

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\theta).$$

## Proof (Continued)

Rewrite the mixed term:

$$\begin{aligned} \mathbb{E}_{q(\mathbf{x}_\sigma)} [\mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)] &= \int \mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma) d\mathbf{x}_\sigma \\ &= \int \mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \left( \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) d\mathbf{x}_\sigma = \iint p_{\text{data}}(\mathbf{x}) \mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x}) d\mathbf{x} d\mathbf{x}_\sigma \\ &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} [\mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})]. \end{aligned}$$

Substitute and complete the square:

$$\begin{aligned} \mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} [\|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma)\|_2^2 - 2\mathbf{s}_{\theta,\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})] + C_1 \\ &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\theta). \end{aligned}$$

# Denoising Score Matching

## Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\theta,\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 \rightarrow \min_{\theta}$$

The optimal MSE predictor is the posterior mean of the regression target:

$$\mathbf{s}_{\theta^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} [\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})]$$

## Marginal Score via Conditional Scores

$$\begin{aligned} \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) &= \frac{\nabla_{\mathbf{x}_\sigma} \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}}{q(\mathbf{x}_\sigma)} \\ &= \int \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)} d\mathbf{x} \\ &= \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} [\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})] \end{aligned}$$

**Therefore:**  $\mathbf{s}_{\theta^*,\sigma}(\mathbf{x}_\sigma) = \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$  - the optimal model learns the true **marginal** score!

# Tweedie's Formula

Let  $\mathbf{x} \sim p_{\text{data}}$ ,  $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$  be independent, with  $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$  and  $\sigma > 0$ .

Using the posterior-mean identity from the previous slide:

$$\begin{aligned}\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) &= \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[ \frac{\mathbf{x} - \mathbf{x}_\sigma}{\sigma^2} \right] \\ &= \frac{\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] - \mathbf{x}_\sigma}{\sigma^2}.\end{aligned}$$

## Tweedie's Formula

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

## Noise Prediction

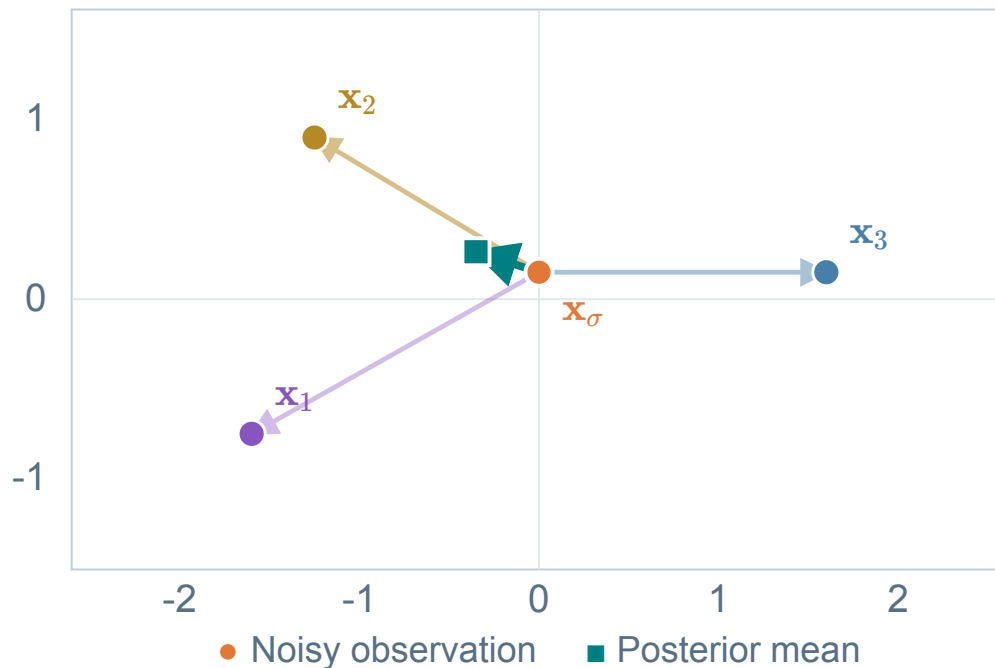
$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_\sigma] = -\sigma \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

Both conditional means are MSE-optimal predictions given the noisy observation.

# From Conditional Scores to Denoising

Toy data: three equally likely clean points  $\mathbf{x}_i$ , with  $\mathbf{x}_\sigma = \mathbf{x} + \sigma\epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$  and  $\sigma = 1$ .

Fixed noisy observation between the clean-data modes.



## Conditional scores

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma | \mathbf{x}_i) = (\mathbf{x}_i - \mathbf{x}_\sigma) / \sigma^2$$

## Posterior weights

$$q(\mathbf{x}_1 | \mathbf{x}_\sigma)$$

22.9%

$$q(\mathbf{x}_2 | \mathbf{x}_\sigma)$$

42.7%

$$q(\mathbf{x}_3 | \mathbf{x}_\sigma)$$

34.4%

## Marginal score = posterior-weighted average

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i q(\mathbf{x}_i | \mathbf{x}_\sigma) \frac{\mathbf{x}_i - \mathbf{x}_\sigma}{\sigma^2}$$

## Tweedie: from the score to the clean-data mean

$$\mathbb{E}[\mathbf{x} | \mathbf{x}_\sigma] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$$

Posterior mean: (-0.35, 0.26)

The MSE-optimal denoised prediction can lie between the clean-data modes.

# Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \rightarrow \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_{\sigma})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_{\sigma}) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_{\sigma})\|_2^2 \rightarrow \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_{\sigma}|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_{\sigma}) - \nabla_{\mathbf{x}_{\sigma}} \log q(\mathbf{x}_{\sigma}|\mathbf{x})\|_2^2 \rightarrow \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \rightarrow \min_{\boldsymbol{\theta}}$$

## Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

# Outline

## 01 Score Matching

Denoising Score Matching

Noise-Conditioned Score Network (NCSN)

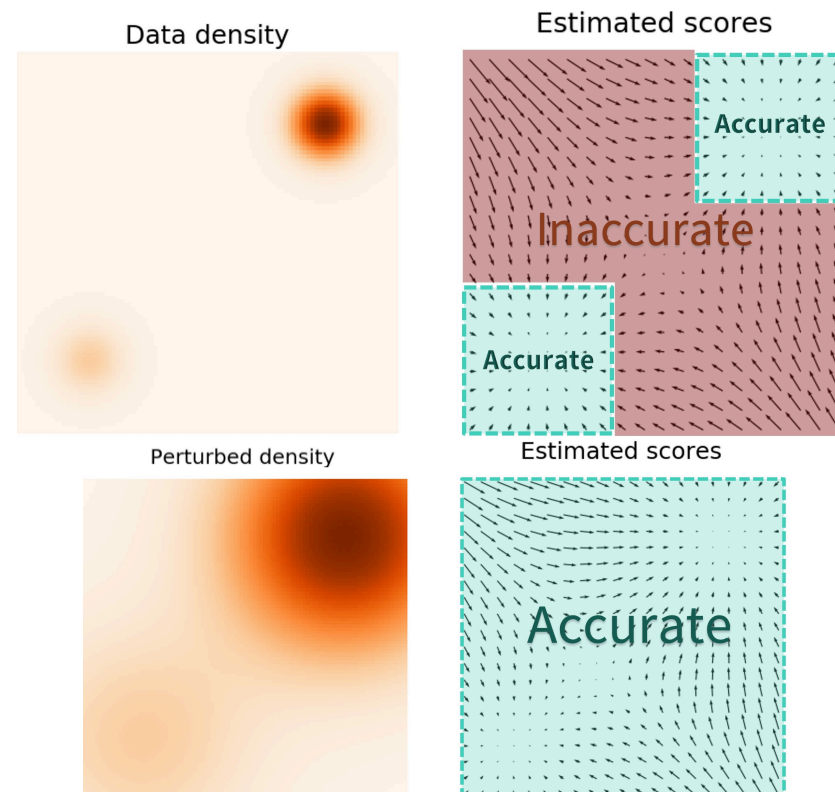
## 02 Forward Gaussian Diffusion Process

# Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0, \mathbf{I})} \left\| \mathbf{s}_{\theta, \sigma}(\mathbf{x} + \sigma \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \rightarrow \min_{\theta}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\theta, \sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l$$

- For **small**  $\sigma$ ,  $\mathbf{s}_{\theta, \sigma}(\mathbf{x})$  becomes inaccurate and Langevin dynamics fails to traverse modes.
- For **large**  $\sigma$ , robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted.

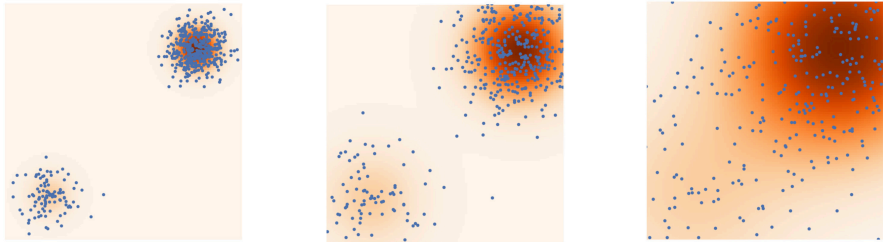


# Noise-Conditioned Score Network (NCSN)

- Specify a sequence of noise levels:  $\sigma_1 < \sigma_2 < \dots < \sigma_T$ .
- Perturb each data point with different noise levels:  $\mathbf{x}_t = \mathbf{x} + \sigma_t \epsilon$ , so  $\mathbf{x}_t \sim q(\mathbf{x}_t)$ .
- Choose  $\sigma_1, \sigma_T$  such that:

$$q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$\sigma_1 < \sigma_2 < \sigma_3$





# Noise-Conditioned Score Network (NCSN)

Train the denoising score function  $\mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t)$  for each noise level  $\sigma_t$  using a unified weighted objective:

$$\sum_{t=1}^T \sigma_t^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \|\mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x})\|_2^2 \rightarrow \min_{\theta}$$

Here,  $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2} = -\frac{\epsilon}{\sigma_t}$

## Training

1. Sample  $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ ,  $t \sim U\{1, T\}$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .
2. Compute noisy image  $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \epsilon$ .
3. Compute loss  $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t) + \frac{\epsilon}{\sigma_t} \right\|^2$ .
4. Update  $\theta$  via stochastic gradient descent.

How do we sample from such a model?

# Noise-Conditioned Score Network (NCSN)

## Sampling (Annealed Langevin Dynamics)

1. Sample  $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$ .
2. Update  $\mathbf{x}_t^l$  via  $L$  steps of Langevin dynamics at each noise level  $\sigma_t$ :

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \boldsymbol{\epsilon}_t^l.$$

3. Update  $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$  and proceed to the next noise level  $\sigma_{t-1}$ .



# Outline

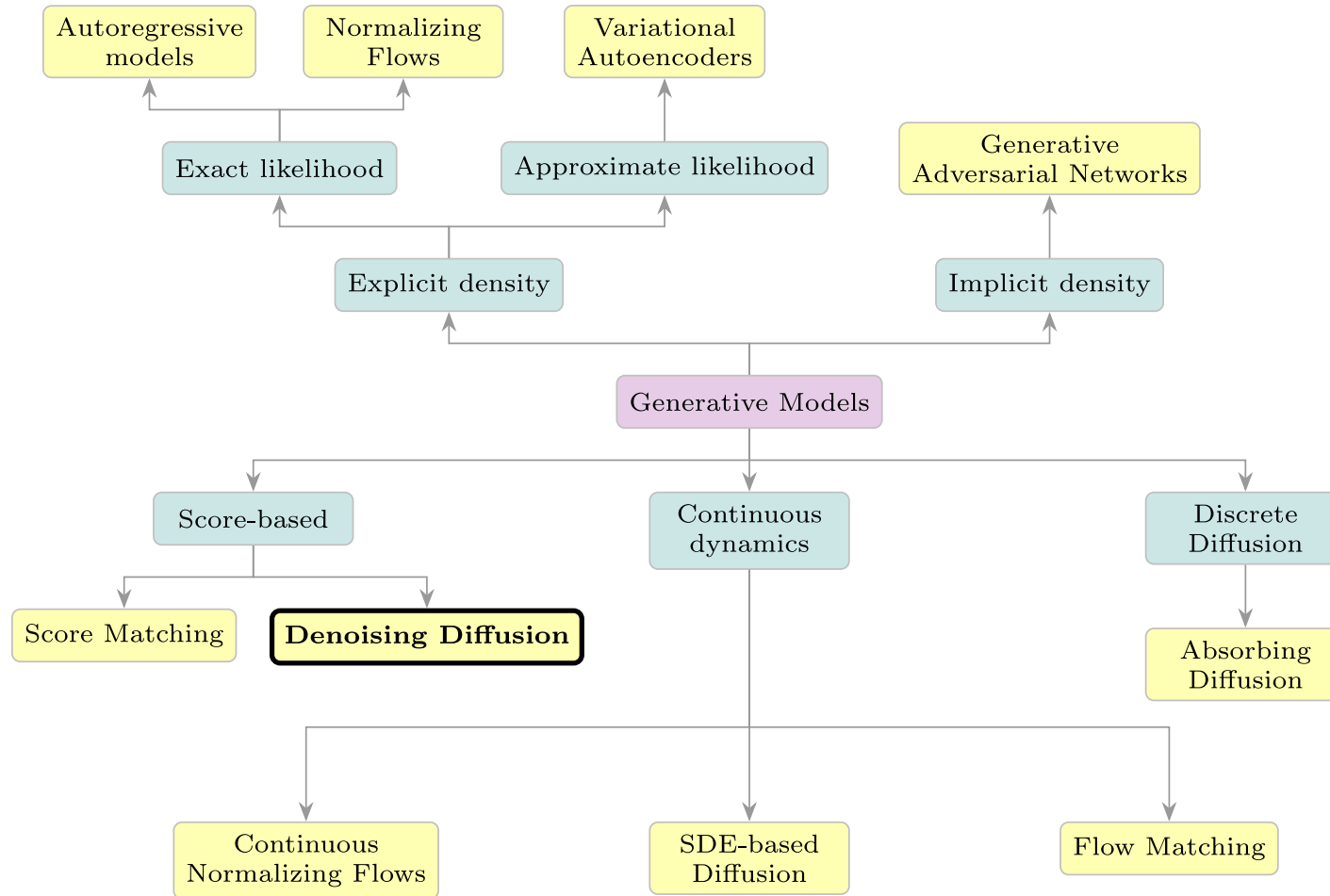
## 01 Score Matching

Denoising Score Matching

Noise-Conditioned Score Network (NCSN)

## 02 Forward Gaussian Diffusion Process

# Generative Models Taxonomy



# Forward Gaussian Diffusion Process

Let  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ,  $\beta_t \ll 1$ . Define a Markov chain:

$$\begin{aligned}\mathbf{x}_t &= \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}) \\ q(\mathbf{x}_t | \mathbf{x}_{t-1}) &= \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})\end{aligned}$$

## Langevin Dynamics

$$\begin{aligned}\mathbf{x}_{l+1} &= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\theta}(\mathbf{x}_l) + \sqrt{\eta} \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}) \\ \mathbf{x}_t &= \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t \\ &\approx \left(1 - \frac{\beta_t}{2}\right) \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t \\ &= \mathbf{x}_{t-1} + \frac{\beta_t}{2} (-\mathbf{x}_{t-1}) + \sqrt{\beta_t} \boldsymbol{\epsilon}_t\end{aligned}$$

- $\beta_t = \eta$
- $\nabla_{\mathbf{x}_{t-1}} \log p_{\theta}(\mathbf{x}_{t-1}) = -\mathbf{x}_{t-1} = \nabla_{\mathbf{x}_{t-1}} \log \mathcal{N}(0, \mathbf{I})$

# Forward Gaussian Diffusion Process

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

## Statement 1

Let  $\alpha_t = 1 - \beta_t$  and  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s = \prod_{s=1}^t (1 - \beta_s)$ . Then

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

Thus, samples at any timestep  $t$  can be generated directly from  $\mathbf{x}_0$

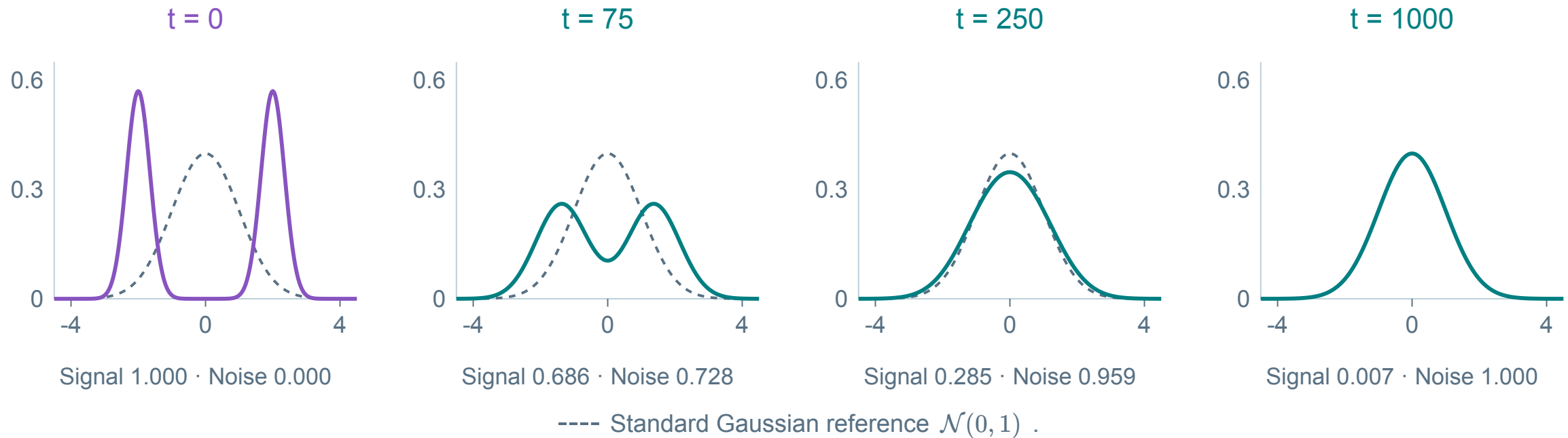
$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_t \\ &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon}_{t-1}) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_t \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + (\sqrt{\alpha_t (1 - \alpha_{t-1})} \boldsymbol{\epsilon}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_t) \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \boldsymbol{\epsilon}'_t \\ &= \dots = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}) \end{aligned}$$

# Forward Gaussian Diffusion Process

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N} \left( \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I} \right);$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N} \left( \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I} \right)$$

$$p_{\text{data}}(x_0) = \frac{1}{2} \mathcal{N}(-2, 0.35^2) + \frac{1}{2} \mathcal{N}(2, 0.35^2), \quad \beta_t = 0.01$$



The plots show marginal densities; the kernels above are conditional.

# Forward Gaussian Diffusion Process

## Statement 2

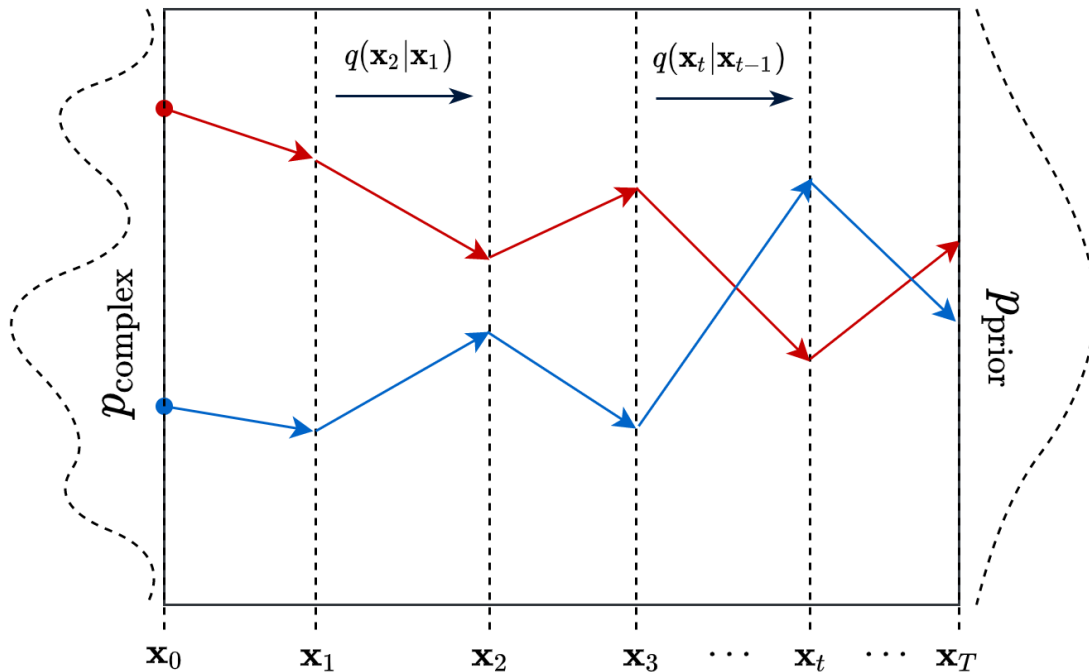
Applying the Markov chain to any distribution  $p_{\text{data}}(\mathbf{x})$  yields  $\mathbf{x}_\infty \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ , the **stationary** (limiting) distribution:

$$\begin{aligned} p_\infty(\mathbf{x}) &= \int q(\mathbf{x}|\mathbf{x}')p_\infty(\mathbf{x}')d\mathbf{x}' \\ p_\infty(\mathbf{x}) &= \int q(\mathbf{x}_\infty|\mathbf{x}_0)p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0 \\ &\approx \mathcal{N}(0, \mathbf{I}) \int p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0 = \mathcal{N}(0, \mathbf{I}) \end{aligned}$$

**Note:** This holds iff  $\bar{\alpha}_t \rightarrow 0$ , i.e.,  $\sum_{t=1}^{\infty} \beta_t = +\infty$ .

# Forward Gaussian Diffusion Process

**Diffusion** describes the migration of particles from regions of high density to those of low density.



1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \epsilon_t,$   
 $\epsilon_t \sim \mathcal{N}(0, \mathbf{I}), t \geq 1$
3. After  $T \gg 1$  steps:  $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

If this process can be reversed, we can sample from  $p_{\text{data}}(\mathbf{x})$  by starting from noise  $p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ . Our goal now becomes inverting this diffusion.

# Denoising Score Matching

## NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

## Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{1 - \bar{\alpha}_t}$$

## Theorem (Denoising Score Matching)

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_t)} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)\|_2^2 \\ &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta}) \end{aligned}$$

**Note:** Annealed Langevin dynamics applies to diffusion, too.

NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

# Summary

- Score matching minimizes Fisher divergence to estimate the score function.
- Denoising score matching makes the score objective estimable using corrupted samples and the known corruption kernel.
- Tweedie's formula converts the score into MSE-optimal clean-data and noise predictions.
- Noise-Conditioned Score Networks learn scores at multiple noise levels and sample with annealed Langevin dynamics.
- The forward Gaussian diffusion process admits a closed-form conditional distribution for any timestep.
- With a suitable noise schedule, the forward process converges to standard Gaussian noise; denoising score matching also applies to these diffusion marginals.